

Pronob Kumar Barman

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EDUCATION

University of Maryland, Baltimore County

PhD in Information Systems, CGPA: 3.85/4.00

01/2022 - 09/2025 (Anticipated)

Research Focus: Generative Models, NLP, Recommendation Systems, UX

University of Kentucky

MS in Statistics, CGPA: 3.85/4.00

08/2019 - 05/2021

Advanced Theoretical Focus: Probability Theory, Statistical Inference, Experimental Design

University of Dhaka

MS in Statistics, CGPA: 3.46/4.00

10/2015 - 01/2017

BS in Statistics, CGPA: 3.46/4.00

01/2011 - 08/2015

PROFESSIONAL EXPERIENCE

University of Maryland, Baltimore County

Teaching Assistant, Information Systems

01/2022 – Present

- Guided graduate students in applying **machine learning techniques** to real-world projects, enhancing **model accuracy** and **project outcomes**.
- Led sessions on **deep learning architectures** and **optimization**, increasing student **understanding by 30%** through hands-on demonstrations.

Technuf LLC

Research Scientist, Intern

06/2024 – 08/2024

- Assessed **economic viability** for Aphelia Student Connect, identifying revenue streams and reducing breakeven time by **30%**.
- Integrated **Generative AI** into **SIEM/SOC systems**, automating threat detection and reducing response times by **40%**.

Data Scientist, Intern

08/2022 – 12/2022

- Optimized **SQL queries** in PostgreSQL to process **200K+ records**, enhancing **operational efficiency** for NY Medicare and Medicaid analysis.
- Built **PowerBI dashboards** and led **Flutter automation testing**, accelerating issue resolution and delivering insights.

University of Kentucky

Teaching Assistant, Department of Statistics

08/2019 – 05/2021

- Assisted students in **statistical methods and causal inference** for real-world projects, increasing student **performance by 20%**.
- Taught **statistical software** and **database languages**, boosting **student comprehension** through applied examples.

Bangladesh Bank

Deputy Director, Chief Economist's Unit

04/2018 – 12/2021

- Directed **economic policy research**, improving national economic models by **65%**, which influenced financial policy and increased **financial stability**.

SKILLS

- **Programming Languages:** Python, R, SQL, SAS, Java, JavaScript, Matlab
- **Data Science Libraries:** NumPy, SciPy, Pandas, NLTK, SymPy, IPython, PyTables, StatsModels, NetworkX, Mayavi
- **ML Tools:** TensorFlow, PyTorch, Keras, Scikit-learn
- **Generative AI:** LangChain, OpenAI API, Hugging Face, Pinecone, ChromaDB, MongoDB
- **Code Debugging:** Profiling tools (e.g., cProfile, Py-Spy)
- **Data Manipulation:** Hadoop, Spark, AWS Cloud, Oracle SQL, PostgreSQL
- **Data Visualization:** Matplotlib, Seaborn, Plotly, ggplot2, Tableau, Power BI
- **Tools:** ConnectWise SIEM & SOC, Git, SPSS, L^AT_EX, Visual Studio IDE, Microsoft Project, Teams, Outlook, Qualtrics
- **Soft Skills:** Leadership, Clear Communication, Strategic Problem-Solving

PHD RESEARCH PROJECT

Online Healthcare Support Group Formation and Recommendation System

- Developed an automated support group formation system using two novel generative models, enhancing **user-group coherence** and **engagement** by 60%.
- Built recommendation systems for online health forum users using **topic modeling** techniques, improving **personalized support group suggestions** by 75%.
- Led a **user experience survey** to assess benefits, features, and privacy, guiding platform improvements based on user feedback.
- Released an [R package for the gSTM model](#), facilitating adoption by researchers and developers.